# INFANT SEPARATION IN MONKEYS:

# STUDIES ON SOCIAL FIGURES OTHER THAN THE MOTHER

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## INTRODUCTION

Nearly twenty years ago, it was first reported that maternal separation in infant monkeys produced effects that bore a strong resemblance to those reported following early loss of the mother in human infants (Seay, Hansen, and Harlow, 1962; Jensen and Tolman, 1962). These findings stimulated a large and continuing research effort into the effects of maternal separation in nonhuman primates (see Mineka and Suomi, 1978 for a review). In most studies, investigators have focused exclusively on the two animals in question: The infant and the mother. However, we feel that much useful information can be gleaned from separation studies in which responses by or to a social figure other than the infant or mother are examined. In each of the three sections of this chapter, we will consider a separate class of such studies, and what each can tell us about the infant monkey's relationship with its mother and other animals. These three classes of studies are: 1) those in which the infant has access to another adult female or "aunt" during the separation period; 2) those in which the infant is separated from an artificial maternal substitute (surrogate) upon which it has been reared; and 3) those in which the response of other animals to mother-infant separation is assessed.

The principal value of the first two classes of studies lies in what they can disclose about the determinants of mother-infant attachment. In the first case, when a substitute female aunt is provided to the infant during separation, she will resemble the mother in some respects and



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differ from her in others. Therefore, we can examine the specificity of the infant's relationship with its mother, and begin to determine what aspects of the mother evoke certain infant responses. Similarly, in the second case, an artificial mother will resemble the real mother in some respects and differ from her in others. By examining the degree to which the infant shows attachment to the surrogate, as measured by the magnitude of the responses to surrogate separation, we can isolate maternal features necessary for the expression of normal filial attachment.

The class of studies that we will consider in the third section of this chapter are of interest because they address the question of infant social relationships within a broader framework. Under natural conditions, in most primate species, the mother and her offspring are part of a larger social network. Although the mother is the most responsive of adults to her young, other group members are attracted to and interact with infants (Hrdy, 1976; McKenna, 1979; Redican, 1976). The reactions of these other animals vary with such factors as their age and sex (Rosenblum, 1972). Thus, the mother's attentiveness to her young might be considered as merely the extreme case of the attentiveness shown by other group members. The third section of this chapter addresses this issue by presenting data on the reaction of adults other than the mother to infant separation procedures.

Throughout this chapter physiological as well as behavioral responses to separation will be reviewed. Only recently have such physiological effects been documented. For instance, Breese et al. (1973) found that mother-infant separation in rhesus macaques produced enzymatic activity indicative of increased sympathetic action and an increase in the brain serotonin levels of infants. Other experimenters have reported changes in sleep patterns, heart rate, and body temperature in separated pigtail macaque infants (Reite and Short, 1978; Reite, Short, Kaufman, Stynes, and Pauley, 1978a). Also, separation procedures have been found to increase pituitary-adrenal activity in both rhesus macaques and squirrel monkeys (Hill, McCormack, and Mason, 1973; Smotherman, Hunt, McGinnis, and Levine, 1979; Mendoza, Smotherman, Miner, Kaplan, and Levine, 1978). We will chiefly be concerned with this latter physiological response. Pituitary-adrenal activity is known to increase following numerous stressful events (Hanson, Larson, and Snowdon, 1976; Coover, Goldman, and Levine, 1971; Friedman and Ader, 1967). Moreover, it appears to be particularly sensitive to psychological, as opposed to physical, stressors and to reflect the organism's state of general arousal (J. Hennessy and Levine, 1979). For these reasons, pituitary-adrenal activity appears to be a particularly suitable measure of the stress of mother-infant separation. Throughout our discussion we will focus primarily on studies using squirrel monkeys *(Saimiri sciureus)* as subjects. These South American primates evidence a strong mother-infant relationship and have been the subject of much recent research in this area (e.g., Kaplan, Cubicciotti, and Redican, 1977, 1979; Mendoza et al., 1978).

## THE EFFECTS OF FAMILIAR ANIMALS UPON THE INFANT'S RESPONSE TO MATERNAL SEPARATION

One of the more crucial factors in the primate infant's response to separation from the mother involves the social environment during the separation experience and the extent to which that environment can satisfy the infant's emotional and physical needs in the mother's absence. Research on the development of both human (Bowlby, 1973) and nonhuman primates (Rosenblum, 1971a) indicates that the infant's response to temporary or sustained loss of the mother is very much dependent upon the infant's experience with multiple maternal figures before maternal loss and upon the availability of these caregiving figures after separation. These influences are probably best illustrated in the work of Rosenblum and Kaufman comparing the separation response of bonnet and pigtail monkey infants (Kaufman and Rosenblum, 1967a, b; Rosenblum and Kaufman, 1968).

Bonnet and pigtail macaques differ greatly in social structure and mother-infant interaction. In particular, it has been reported that bonnet monkeys are far less aggressive and maintain far smaller interindividual distances than do pigtail monkeys (Kaufman and Rosenblum, 1967b; Rosenblum, 1971b). The infant's interactions with its mother and other group members are consonant with these social structure differences; bonnet infants from birth are touched and handled by other group females, while pigtail infants remain with the biological mother and spatially separated from other group members (Rosenblum and Kaufman, 1968; Rosenblum, 1971b). Therefore, bonnet infants are reared in a social environment consisting of the mother--the primary focus of attachment --and, in addition, several other affiliative figures. Pigtail infants in contrast develop in a much more punitive, restrictive, and exclusive maternal environment (Rosenblum, 1971b). Kaufman and Rosenblum reported that the response of bonnet and pigtail infants to maternal separation was correspondingly quite different. Infants of both became immediately agitated upon removal of the mother from the group. However, bonnet infants quickly established contact with the familiar adults of the group and showed no great disruption in normal behavioral patterns during the several weeks of separation (Rosenblum and Kaufman, 1968). The pigtail infants, in contrast, showed a longer initial period of intense agitation, lasting usually a complete day, and then lapsed into a physical and behavioral depression characterized by a hunched posture and greatly decreased spontaneous behavior (Kaufman and Rosenblum, 1967a, b). Thus, when other group members are accessible to the infant during maternal separation, removal of the mother for an infant monkey reared in a social environment with alternative, albeit secondary, attachment figures appears to be far less traumatic than for an infant reared in an environment with only a single maternal figure.

The importance of alternative maternal figures has also been demonstrated with research on a New World primate, the squirrel monkey.

For the first few months of its life, the squirrel monkey clings almost constantly to the mother's back (Rosenblum, 1968, 1974), as shown in Figure 1. Under some circumstances squirrel monkey infants, which are often subject to "aunting" attention from other animals in the social group (Hunt, Gamache, and Lockard, 1978; Rosenblum, 1968), may form special long-term maternal-like relationships with other females besides the mother (Rosenblum, 1971a). This aspect of squirrel monkey social organization was employed by Rosenblum (1971a) to test further the impact of single versus multiple maternal figures during development upon the infant's response to maternal separation. Two types of mother-infant groups were formed for his study. The "monomatric" group contained only mothers and their infants while the "polymatric" groups included several prospective aunting females in addition to the mother-infant pairs. When an infant was 4-5 months of age, the mother was removed for several days while the infant remained in the social group. The results of the experiment indicated that even within the first day of the separation period the infants that were aunted in the polymatric groups showed virtually no behavioral distress. In contrast, the separated infants in the monomatric groups, that did not receive substitute maternal care, were highly agitated for several days (Rosenblum, 1971a). These data on squirrel monkey infants complemented Rosenblum's (Rosenblum and Kaufman, 1968) previous research on bonnet and pigtail monkeys and showed that even within a single species variations in substitute maternal care can have a dramatic influence on infant separation responses.

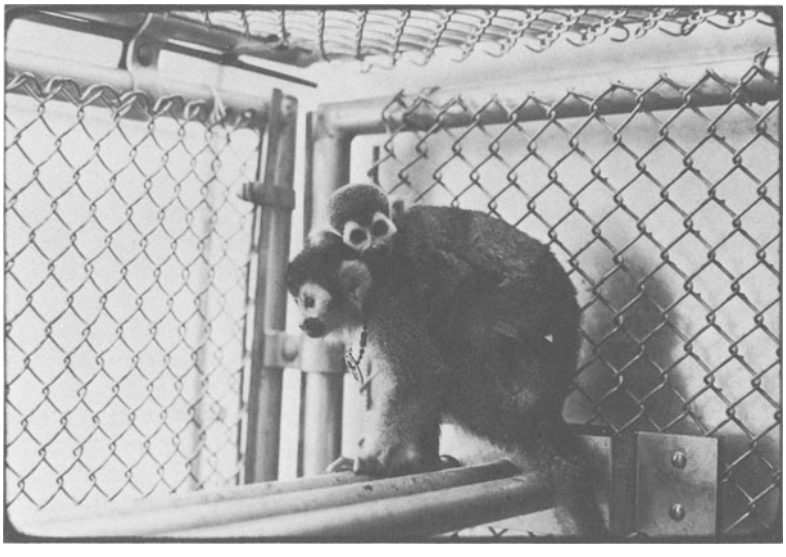


FIGURE 1

Mother and Infant Squirrel Monkey, with Infant Clinging to Mother's Dorsal Surface in Characteristic Position

More recently, a group of our colleagues has examined the effects of aunting upon the adrenocorticoid response of the infant squirrel monkey to brief separation. In an initial study (Mendoza et al., 1978), it was found that infant squirrel monkeys living only with the mother exhibited a marked elevation of plasma cortisol levels when separated from her and placed alone in an unfamiliar environment for 30 minutes. Control conditions involving handling but no mother-infant separation showed that this pituitary-adrenal activation was not due to merely the disturbance of the separation procedure. Subsequently, Coe, Mendoza, Smotherman and Levine (1978) examined the cortisol response to separation under conditions in which familiar social stimuli were available during the mother's absence. Infants left in the home cage with familiar social partners during the 30-minute separation again showed the significant increase in plasma cortisol levels. Notably, this physiological stress response was found despite the fact that 3 of the 4 infants were clearly aunted (i.e., were observed riding on a familiar female's back) during the separation period. Thus, although aunting appears capable of reducing behavioral signs of distress, this study indicated that a major physiological response to stress was not similarly ameliorated.

Because these results were intriguing to us, Vogt, Lowe and Levine (in prep. a) examined the question in greater detail and directed particular attention to the behavior of the infant following the mother's removal. This study utilized a social environment optimal for aunting of the infants during both normative development and maternal separation. The animals of this study were formed into 4 long-standing groups so that infants were born and developed within relatively stable social environments. Each group of 5-8 adult females included several potential aunts, some of which had lost that season's infant near birth and were strongly motivated to act maternally. Extensive behavioral observations of infant development and social interaction were obtained under normative conditions and were compared with the infant's behavioral patterns during the maternal separations. In one of the separation conditions (Mother Removal), the mother was taken to a cage in another room and the infant was left behind in the social group. In a second condition (Infant Removal), the mother was immediately returned to the social group after separation while the infant was housed in the small cage in the separate room. For each of the mother-infant pairs tested, behavioral observations of the infant were recorded during each 30-minute experimental test. At the end of these sessions, as well as under undisturbed basal conditions, blood samples for analysis of plasma cortisol concentrations were obtained.

It was found that the infants' cortisol values after both of the separation conditons were significantly elevated above basal values. Moreover, the pituitary-adrenal activation of the infants was equivalently great whether they were left in the home cage with familiar animals or were removed from the group. The behavioral data, however, revealed that the infants' activity within the social group was little changed by separation and removal of the mother. In Figure 2 the data recorded

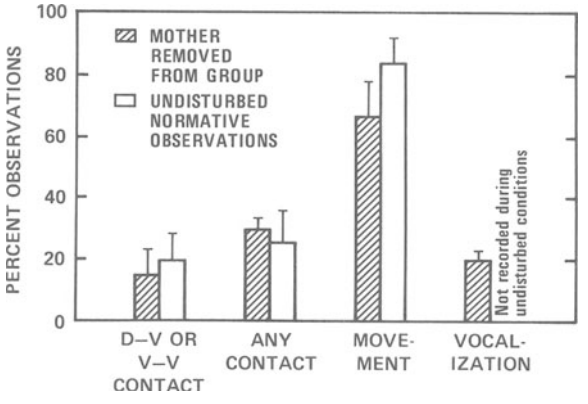



FIGURE 2

Behavioral activity of group-living infants during maternal separation or during normative observations recorded in the same week as the separation condition (from Vogt et al., in prep.).

during the *Mother Removal* condition are compared with the normative observations. These normative data were gathered before any manipulation on four mornings of the week in which the *Mother Removal* test was used with a particular infant. As can be seen, movement and contact scores were not changed during the mother's removal. Vocalizations were not systematically scored during normative conditions and therefore were not compared with the *Mother Removal* data. Although the vocalization score of about 20 percent during the mother's removal could reflect some increase in agitation over normative conditions, the elevation was not great. This appears to be the case because the infants' vocalization levels were much higher during removal from the group in more than 75 percent of the observations.

Because the infants tested in the Vogt et al. study were approximately 4-6 weeks older than the infants of the Coe et al. study, the rather low percentages presented in Figure 2 for time clinging to a mother or aunt during normative and separation conditions were expected. Moreover, since only 3 of the 9 infants in the Vogt et al. study were aunted for substantial portions (22-65%) of the separation period, it appears that the mere presence of the familiar social and physical environment was at least as important as contact with a substitute maternal figure in the alleviation of behavioral agitation during the mother's absence. Overall the results of this study indicate that pituitary-adrenal activation can occur under separation conditions in which the behavior of the infant closely resembles that observed following no disturbance. Further, the Coe et al. and Vogt et al. studies raise the somewhat discomforting possibility that infants may experience clear physiological stress reactions to manipulations such as maternal separa-

tion even when behavior undergoes no obvious change. In any event, future investigators might do well to consider what response systems have and have not been monitored when speaking of the infant's "separation distress" during various conditions of maternal separation.

## SEPARATION RESPONSES IN SURROGATE-REARED INFANTS

The rearing of monkeys on artificial mothers was first introduced by Harry Harlow and his associates at the University of Wisconsin. In their classic experiments (Harlow, 1958; Harlow and Zimmerman, 1959), infant rhesus macaques that were removed from their mothers at birth and raised exclusively on terry cloth-covered wooden cylinders were found to cling and other wise respond to the inanimate surrogates as they would to their own mothers. However, if the surrogate's surface was of wire mesh rather than terry cloth, infants spent less time on the surrogate and generally showed less attraction to it. These results indicated that tactual stimulation or "contact comfort" was a critical factor in the development of the infant's attachment to its rearing figure. Features other than tactual stimulation, such as the association of the surrogate with nursing, its warmth, facial features, and even the presence or absence of a head (Harlow and Zimmerman, 1959; Harlow and Suomi, 1970) each had a secondary, if any, effect on the infant's affectional responsiveness to its surrogate. Recent observations have extended these findings by demonstrating that infant monkeys will show filial attachment to objects as diverse as live dogs and toy hobby horses as long as they provide a soft furry surface upon which to cling (Mason, 1979; Mason and Kenney, 1974). The attachment of infant monkeys to surrogate mothers is not limited to Old World or macaque species. Infant squirrel monkeys can be raised readily on cylindrical, cloth-covered surrogates (Kaplan, 1974). The young squirrel monkeys cling to and nurse from surrogates in the species-typical fashion (see Figure 3), and by 4 weeks of age can recognize and prefer their own rearing surrogate to other similar surrogate mothers (Kaplan and Russell, 1974).

These and other studies on surrogate rearing have added much to our understanding of the infant primate's relationship to its mother. The surrogate model is particularly useful because it offers a means of completely controlling the reactions of the maternal figure. Hence, specific features of the surrogate can be systematically varied and their effects on particular infant responses can be examined without the potential confounding influence that the mother's own responsiveness to the infant provides. Of course, the surrogate model, like all models, is a simplification of the actual phenomenon. Real mothers and infants are continually interacting with each other, and almost certainly these interactions have some effect on the attraction and attachment that the infant develops for the mother. Thus, as filial attachment is studied in greater detail, one would expect to find discrepancies between mother-reared and surrogate-reared infants. When this occurs, the surrogate model becomes useful in another way. In addition to the original question,

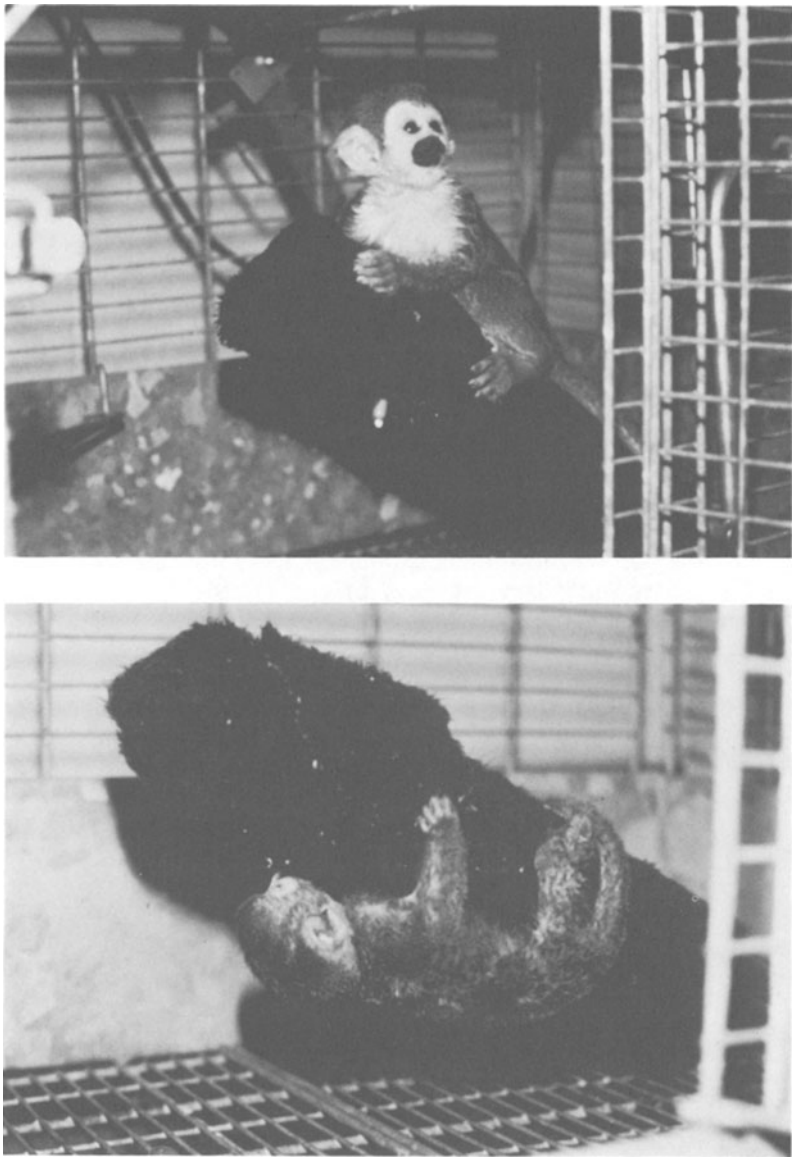


FIGURE 3

Surrogate-reared squirrel monkeys. Upper photograph shows an infant riding dorsally on its surrogate. In the lower photograph an infant has moved ventrally to nurse. The nipple protruding from the base of the surrogate is attached to a plastic bottle containing infant formula.
(Photographs courtesy of J. Kaplan)

"What is similar between mother and surrogate (e.g., contact comfort) that leads to the similarity in the infant's attachment to the two rearing figures?", we can also ask, "What is different between mother and surrogate that leads to the difference in the infant's attachment to these two figures?".

Recent evidence suggests that we have reached just such a juncture. Studies with both Old and New World primates indicate clear distinctions in the filial attachment of mother-reared and surrogate-reared infants. Interestingly, physiological measures appear to be at least as discriminating as behavioral measures in detecting these differences. Meyer, Novak, Bowman, and Harlow (1975) compared the responses of mother-reared and surrogate plus peer-reared rhesus infants during a 9-week separation from their attachment objects. Overall, the mother-reared group showed more severe behavioral signs of distress (e.g., suppressed levels of play) than did surrogate plus peer-reared infants. Measurements of plasma cortisol showed that, whereas resting levels prior to separation did not differentiate the two groups, mother-reared infants had higher cortisol levels than did surrogate plus peer-reared infants following separation. This finding was primarily due to a difference in the values of the second of two animals tested each day. Thus, the mother-reared but not surrogate plus peer-reared infants showed either a response to the disturbance of the first animal or an anticipatory response to the testing procedures, which was signaled by the disturbance of the first animal. In another study (Reite, Short, and Seiler, 1978b) it was found that during a 4-day separation from their surrogates, infant pigtail macaques showed changes in behavior, sleep patterns, heart rate, and body temperature that were less pronounced than previously observed in mother-reared pigtail infants (Reite and Short, 1978; Reite et al., 1978a). Overall, the surrogate-reared infants evidenced a possibly attenuated "agitation" stage and a virtual absence of the subsequent "depression" stage of separation that are characteristically observed in mother-reared pigtails.

In the squirrel monkey, an initial investigation of physiological (pituitary-adrenal) responsiveness to separation in surrogate-reared and mother-reared infants presented a somewhat different picture (Mendoza et al., 1978). Infants taken from their surrogates and placed in a different cage in an unfamiliar room for 30 minutes showed a clear elevation of plasma cortisol levels that was as large as that shown by mother-reared infants under comparable conditions. The disturbance of the separation procedure had no effect as indicated by a lack of cortisol response when the infant and rearing figure were separated but then immediately reunited. Therefore, the results of this study suggested that the adrenocorticoid response to separation from the maternal object was equivalent in surrogate- and mother-reared squirrel monkeys. However, since Mendoza et al. did not examine the effect on cortisol levels of the unfamiliar separation environment, it was possible that the response of either or both of the two groups of infants was at least partially

stimulated by the exposure to the unfamiliar surroundings. As noted in the previous section of this chapter, later experiments (Coe et al., 1978; Vogt, Lowe, and Levine, in prep. a) found that mother-reared infants showed as large a cortisol response to separation when the mother was taken from the home cage as they did when the infant was removed to an unfamiliar environment. Thus, it appears, at least for mother-reared infants, that separation alone can account for the pituitary-adrenal activation observed following these separation procedures.

In surrogate-reared squirrel monkeys, the relative contributions of separation and novelty to the cortisol response was examined by Hennessy, Kaplan, Mendoza, Lowe, and Levine (1979). Infants were tested under the five conditions shown in Figure 4. In the first (BASAL) condition, each infant was undisturbed and a blood sample was rapidly obtained in order to determine resting levels of cortisol. In the SEP.UNF.HAN. condition, each infant was separated, placed in an unfamiliar environment for 30 minutes, and of course, experienced the handling this procedure entailed. The third condition (SEP.FAM.NON HAN.) involved separation, but no exposure to unfamiliar surroundings or handling. The surrogate was simply slipped from the infant's home cage for the 30-minute period. In the fourth condition, (NON SEP.FAM.HAN.) the infant was briefly handled by the experimenter, but there was no further separation or exposure to novelty. In the last condition (SEP. FAM.HAN.) the infant was separated (had the surrogate removed from the cage as in the third condition) and was handled (as in the fourth condition), but was not exposed to novelty.

As is apparent in Figure 4, only the second condition, which involved exposure to novelty, resulted in an elevation of plasma cortisol above resting levels. Separation alone (Condition 3), handling alone (Condition 4) and separation together with handling (Condition 5) each had no effect on plasma cortisol levels. These results, together with those of Coe et al. (1978) and Vogt et al. (in prep. a), suggest that surrogate- and mother-reared squirrel monkeys do in fact differ in their responsiveness to their maternal objects. Whereas separation without exposure to novelty produced a cortisol response in mother-reared infants, this procedure had no observable effect on the cortisol levels of surrogate-reared infants.

As was discussed above, the mother-reared infants of Coe et al. (1978) and Vogt et al. (in prep. a) were left in the presence of other females during the separation period. Since infant squirrel monkeys recognize and prefer their mother to other females (Kaplan, 1977; Kaplan et al., 1977), one could argue that the cortisol responses of mother-reared infants observed by Coe et al. (1978) and Vogt et al. (in prep. a) resulted to some degree from detection that the female to which the infant was clinging or otherwise interacting with was not its mother. That is, the difference in cortisol concentrations observed in mother-reared and surrogate-reared infants might be attributed to the presence or absence of other social figures rather than to a difference in the response to

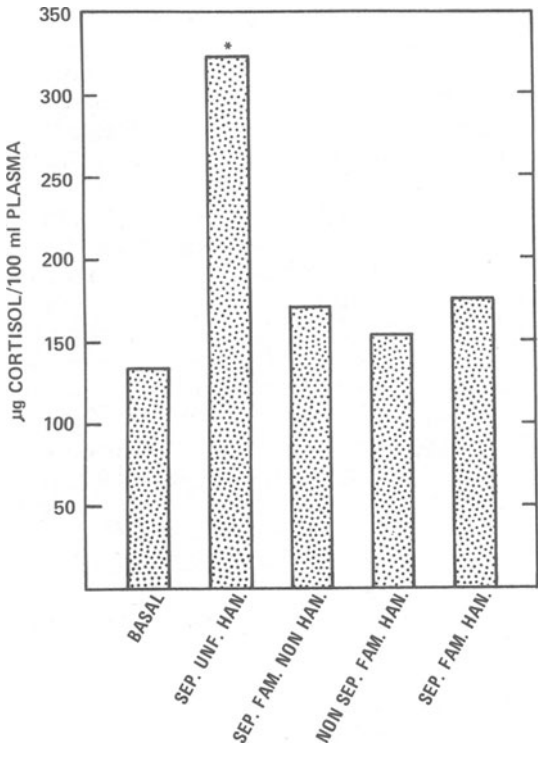



FIGURE 4

Plasma cortisol levels of surrogate-reared squirrel monkeys in five experimental conditions: BASAL; SEP.UNF.HAN. (infant separated into an unfamiliar environment and handled); SEP.FAM.NON HAN. (infant separated, left in familiar environment and handled); NON SEP. FAM.HAN. (infant not separated, left in familiar environment and handled); SEP.FAM.HAN. (infant separated, left in familiar environment and handled). *differs from all other means, p's < 0.01. (from Hennessy et al., 1979)

separation from the attachment object. However, in a second experiment by Hennessy et al. (1979) it was found that separation followed by immediate replacement with discriminable substitute surrogates produced no increase in cortisol levels. Behavioral responses to the familiar versus unfamiliar surrogates (sniffing, time spent on) confirmed that the infants did in fact discriminate between these conditions. Further, considerable amounts of time were spent clinging to each of the surrogates. These results thus point to a difference between mother-reared and surrogate-reared infants in attachment as responsible for the observed difference in cortisol levels during separation.

In this same study, behavioral measures of separation distress were also observed and are illustrated in Figure 5. Measures of vocalizations, movement, and environmental manipulation were obtained during the 30 minutes following complete removal of the surrogate from the cage (SEPARATED), removal of the rearing surrogate and replacement with discriminable substitutes that were either clean (CLEAN SUR.) or had been soiled by a peer (PEER SUR.), or removal and immediate replacement of the infant's own surrogate (OWN SUR.). Examination of this figure shows that separation without replacement led to significant increases in vocalizations and movement. Separation, even when followed by replacement with a substitute surrogate, resulted in a depression of environmental manipulation. Thus, in terms of behavior, these infants were responding to separation by showing signs of agitation, just as mother-reared squirrel monkey infants have been observed to do (Jones and Clark, 1973; Kaplan, 1970). If behavior alone had been examined in the Hennessy et al. study, the investigators probably would have concluded that separation from the surrogate and separation from the mother have similar effects. It was only when the physiological measure was examined that a difference in the response of surrogate-reared and mother-reared infants was apparent. The cortisol data indicate that separation from the rearing surrogate is less stressful than is separation from the mother, a conclusion that is entirely consistent with those of Meyer et al. (1975) and Reite et al. (1978b).

One issue that these studies raise is why physiological measures appear to be particularly sensitive to differences in filial attachment, a process generally measured in terms of behavior. In addition, as we noted above, the results also raise the question of what distinction between surrogate and mother accounts for the difference in responsiveness shown to these two rearing figures. Definitive solutions to these problems are not available at this time, but we would like to suggest some hypotheses that appear to fit the available data and could be tested in future studies. With regard to the first of these issues (physiological versus behavioral measures), there is virtually universal agreement that, under most conditions, involuntary separation from the attachment object is a stressful event for the infant. Separation would therefore be expected to affect the physiology of the infant in the same manner as other stressful events, e.g., to induce heightened pituitary-adrenal activity. The infant's behavior, on the other hand, is determined both by its internal state and by specific stimulus features of the environment. Thus, if one is interested in determining whether separation from the rearing figure is stressful for the infant, we would suggest that physiological effects provide a *more direct measure* than do behavioral changes. Moreover, the mother influences the infant not only through her role as the object of emotional attachment, but also by being a critical stimulus for the young monkey. That is, particular stimulus properties of the mother elicit from the infant certain species-typical responses. We feel that it is important to consider this role of the mother as a stimulus figure when drawing inferences about changes in the infant's behavior. Perhaps the mother's role as a stimulus

figure is best illustrated by the clinging response. It is clear that features of the mother elicit clinging. This effect cannot be explained by the mother's status as the attachment object because if a young infant is removed from the mother's back, it will cling to other available objects, such as the hand of the experimenter. Moveover, if one is willing to accept that events which occur after birth are significant factors in the development of attachment, then one is led to conclude that the mother's role as a stimulus figure precedes her role as an attachment object. During the birth process, the infant squirrel monkey has been observed to grasp the mother's fur before completely emerging from the birth canal and within a minute of birth to climb onto the mother's back (Bowden et al., 1967; Takeshita, 1961-1962). The mother is therefore serving as an effective stimulus object for clinging from the moment of birth.

It is suggested here, that in the conventional separation procedure, two important events occur simultaneously. The infant is deprived of its attachment object, and an important aspect of the stimulus situation is altered. The primary effect of removing the attachment object is to produce a heightened state of stress or arousal, as reflected in pituitary-adrenal activity, whereas the primary effect of altering the stimulus

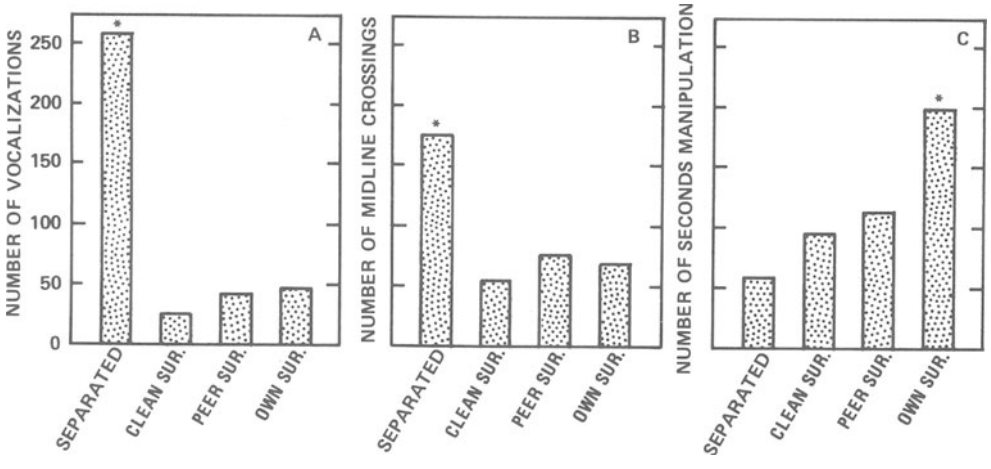



FIGURE 5

Number of vocalizations, midline crossings (activity) and seconds of environmental manipulation by surrogate-reared infants in four experimental conditions: SEPARATED (surrogate removed from home cage); CLEAN SUR. (surrogate removed from home cage and immediately replaced with a clean surrogate); PEER SUR. (surrogate removed from home cage and immediately replaced with the surrogate of a peer); OWN SUR. (surrogate removed and then immediately placed back in the home cage). *differs from all other means, p's 0.05 (A,C), p's 0.01 (B). (from Hennessy et al., 1979)

situation is to elicit species-typical responses, e.g., vocalizing and locomoting. These behaviors may have evolved to occur under these stimulus conditions (absence of mother) because they increase the probability that mother and infant will be reunited. Vocalizing could increase the probability that the mother will approach the infant, and movement could increase the probability that the infant will contact the mother.

Although removal of the attachment object and the maternal stimulus typically occur together, they can be dissociated. In the aunting experiments discussed above (Coe et al., 1978; Vogt et al., in prep.), infants were deprived of their attachment object but important maternal stimuli were replaced with those of the aunt (e.g., her back constituted a suitable stimulus for clinging). The results were as might be predicted; a heightened state of arousal, as indicated by an increase in cortisol levels, but little change in behavior. The complement of this situation appears to have occurred in the surrogate-separation experiment (Hennessy et al., 1979). Because the infant's attachment to its surrogate presumably was not equivalent to the attachment shown by infants to real monkey mothers, removal of the surrogate did not induce the heightened state of arousal or associated increase in cortisol levels. However, removal of the surrogate (when not replaced with a substitute surrogate) did deprive the infant of maternal stimuli. This resulted in an increase in behaviors (vocalizations and locomotion) that are associated with this change in the stimulus situation.

The second issue that needs to be addressed is the difference between mother and surrogate that accounts for the difference in attachment that infants show to these two figures. This question, like the first, must be answered more with speculation than with a firm data base. To help generate hypotheses, it might be useful to consider more closely the perspective of the infant squirrel monkey. As mentioned above, the infant rides from the moment of birth on the mother's back. It is only beyond the first month of life that the young monkey begins to spend any appreciable amount of time off her (Rosenblum, 1968). The mother, in turn, moves about the environment which tends to highlight to her offspring the distinction between her and the surrounding objects. Further, the mother initiates actions toward the infant and continually reacts to the infant's behavior. Thus for the young squirrel monkey, the most prominent, salient, and stable feature of the constantly changing environment is this thing we call the mother. Moreover, it has been suggested that a contingency is established between infant behaviors and maternal responses that permits the infant its first measure of control over the environment (Watson and Ramey, 1972; Lewis and Goldberg, 1969; Smotherman, Wiener, Mendoza, and Levine, 1976; Wood, Mason, and Kenney, 1979). In light of these occurrences, it is not surprising that the infant begins to recognize the mother and to prefer her to other females. Nor is it surprising that separation from the mother should be stressful for the infant. That is, it is not surprising that the infant begins to meet the common criteria established for attachment.

For the surrogate-reared infant, the situation is somewhat different. Although it spends much of its early life clinging to the surrogate, the surrogate does not command the infant's attention as does a monkey mother. Further, the surrogate may be less discriminable from other environmental features than is a real mother. The surrogate neither moves, initiates actions toward the infant, nor responds to the infant's behavior. Further, because the surrogate is a stationary object firmly affixed to the cage, the infant might not clearly perceive the distinction between surrogate and other environmental features. That is, it may be easier for the experimenter than it is for the monkey to determine just where the surrogate ends and the cage begins. Under these conditions, the surrogate-reared infant might exhibit attachment to familiar aspects of the environment in addition to the surrogate. In other words, the surrogate-reared infant's attachment may be more diffuse and not as clearly limited to the rearing figure as is the attachment of the mother-reared infant. This hypothesis would offer an explanation for the cortisol data obtained with surrogate-reared squirrel monkeys (Mendoza et al., 1978; Hennessy et al., 1979). When the surrogate was simply removed from the cage, the infant did not exhibit physiological signs of arousal because all potential attachment objects were not removed. The home cage and other familiar stimuli in the rearing room remained with the infant. It was only when the infant was taken to a different environment, away from all potential attachment objects, that physiological signs of arousal were observed.

## RESPONSIVENESS OF OTHER ADULTS TO MOTHER-INFANT SEPARATION

As we have emphasized at several points in this paper, the social development of the normal primate infant does not depend exclusively upon one primary attachment figure. Even when clinging at an early age almost constantly to its mother, the infant is frequently interacting with other members of its social group. The final section of this chapter examines the infant as a focus of social interactions in the group and as an animal with age-related characteristics that attract others to it. Several experimental studies are described that test the behavioral and pituitary-adrenal responsiveness of various classes of animals to an infant after removal from its mother. In addition, research on the beginnings of the infant's social responsiveness to animals other than the mother is briefly described.

A newborn monkey has several physical characteristics that distinguish it from other members of its group. Squirrel monkey infants, like infants of all primate species, are physically much smaller and proportioned differently than older conspecifics; the neonates of several other species are in addition distinctively colored and marked compared to adults. These and other salient, unique features of the infant monkey serve to focus social attention upon, and to inhibit aggression toward, the young and dependent monkey. Certain members of the social group

appear to be particularly attracted to infants and their mothers. The extent of this alloparental (Wilson, 1975), aunting (Hinde, 1965), or paternal (Itani, 1959) care varies with several factors, including the species being considered and the sex, reproductive status, and rearing history of the caregiving animal (Hinde, 1965; Sackett, 1970; Lancaster, 1971; Rosenblum, 1971a; Mitchell and Brandt, 1972; Sackett and Ruppenthal, 1974). Generally, females more than males are attracted to infants and will more readily care for them (Hrdy, 1976). However, in primate species showing monogamy and family-group social structure (e.g., the Callitrichids), males are observed to be at least as active as females in caring for and interacting with infants (e.g., Vogt, Carlson, and Menzel, 1978).

To assess more systematically the attraction of various animals to infants, research has been conducted which uses the infant as a stimulus animal. Presentation of the young animal without the presence of the mother allows the testing of affiliative responsiveness without maternal interference or inhibition. For example, Sackett and his colleagues performed a series of experiments (reviewed in Sackett and Ruppenthal, 1974) on the determinants of infant attractiveness in rhesus and pigtail adult females. It was reported that wild-born females, if they had had previous maternal experience, were attracted to infants while lab-born females, regardless of past maternal experience, did not find infants attractive. During the rearing period, monkey mothers prefered their own to other infants, and this preference was heightened considerably when rearing occured in a nonconfined, social group environment. Mothers did not, however, prefer their own infants to other adult females except when the infant was less than one month of age (Sackett and Ruppenthal, 1974).

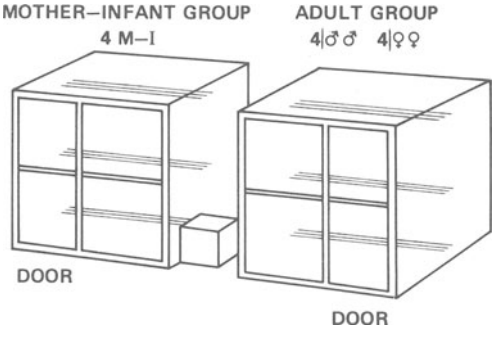
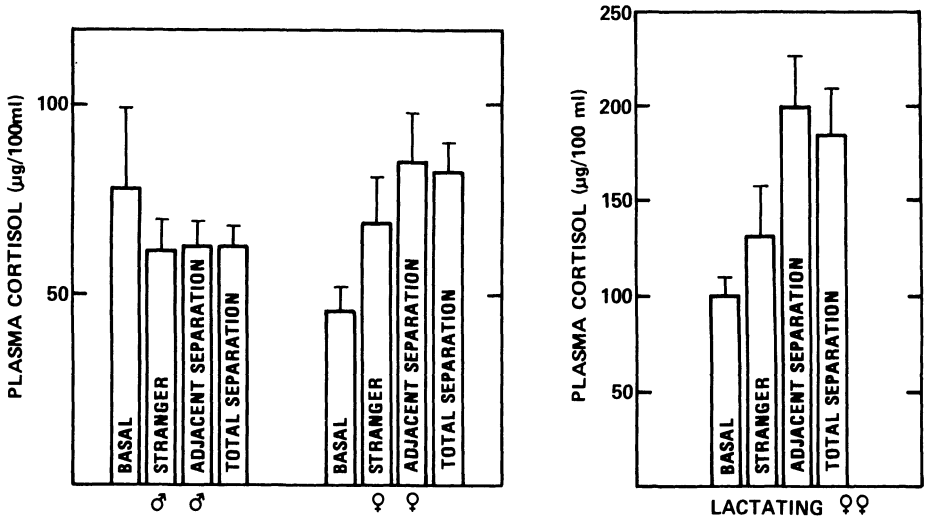



FIGURE 6

Housing conditions for mother-infant and adult groups in Vogt et al. (1980) study. Small cage between two group cages was used as stimulus cage during experimental manipulations, and all cages were constructed of wire mesh.

The research on responsiveness to squirrel monkey infants has tested both males and females of many ages and recently has involved hormonal as well as behavioral measures. Rosenblum (1972) conducted an experiment in which a young (17-24 day old) infant was removed from its mother and introduced into a group of males and females of different ages. After an individual had retrieved or approached the infant, that animal was removed from the group, and the remainder of animals again exposed to the infant. It was reported that mature adult males were the least likely to approach, contact or retrieve the infant, while subadult or preparturient females initiated the greatest number of contacts with the separated infant. Subadult males made slightly more contacts with the infant than did the adult males, while the adult nonpregnant females approached less than subadult or pregnant females but more than subadult males. This relative avoidance of infants by adult squirrel monkey males has also been observed when males are housed in undisturbed laboratory groups with mothers and infants (Rosenblum and Coe, 1977).

Overall, in squirrel monkey groups various types of females are attracted to infants while adult males tend to avoid mother-infant subgroups. However, adult males are by no means insensitive to manipulations of the infant; it is our experience that adult male squirrel monkeys will show protective behavior by physically attacking human caretakers who handle a mother-infant dyad. These informal but quite impressive observations of male protection, plus other anecdotal reports and the research of Rosenblum (1972), led to an experiment assessing the physiological, as well as behavioral, responses of adult squirrel monkeys to mother-infant separation (Vogt, Coe, Lowe, and Levine, 1980). In this study, a group of mother-infant dyads was housed next to a group of adult males and females. As can be seen in Figure 6, there was enough space between the two group cages for a small wire-mesh stimulus cage which was used during the experimental conditions. Although the mothers and infants were physically separated from the adults of the other group, the wire-mesh of the group cages allowed continual visual, auditory, and olfactory communication between the groups. Also, any manipulations of a mother-infant pair were clearly seen and heard by all other animals. The weekly test sessions in this study consisted of four conditions: Adjacent Separation, Total Separation, Stranger, and Basal. In both of the separation conditions, a mother-infant pair was separated, with the mother left in the home cage and the infant placed either in the stimulus cage between the two group cages (Adjacent) or in a small cage in a separate building (Total). The Stranger condition was a control for the social agitation of the separation procedure and involved placing a juvenile squirrel monkey, unfamiliar to any of the animals, into the stimulus cage between the two groups. In the basal condition, animals were undisturbed. Thirty minutes after each of these procedures, blood samples were obtained from the males and females in the adult group and from the nonseparated mothers (lactating females) in the mother-infant group. In addition, behavioral observations were recorded during each 30 minute session; the animals in the adult group were scored for proximity and approach to the stimulus cage.




FIGURES 7A and 7B

Plasma cortisol levels of males and non-lactating females in adult group (A) and lactating females in mother-infant group (B), in response to basal stranger (disturbance control), and two infant separation conditions. (from Vogt et al., 1980)

It was found that females regardless of reproductive state showed an adrenocorticoid response to mother-infant separation. As can be seen in Figure 7A, the females in the adult group and the lactating females in the mother-infant group both exhibited cortisol values after the Adjacent and Total Separation conditions that were significantly elevated over the Basal values. Moreover, the disturbance resulting from merely having an agitated monkey in the stimulus cage cannot account for this response, as the Stranger condition failed to induce significant elevations over Basal levels for either type of female. The males' hormonal values, in contrast, were equivalent across the four conditions (Figure 7B). These data suggest that squirrel monkey females, whether lactating or not, are hormonally more responsive to mother-infant separation than are males.

The behavioral data of this study revealed comparable sex differences, as seen in Figure 8. During the Adjacent Separation sessions, the males totally avoided the stimulus cage, which contained the agitated infant. When the stimulus cage was empty (Total Separation) or occupied by the unfamiliar conspecific (Stranger), the frequency of approaches by the males was significantly greater. The females, on the other hand, revealed a quite different trend; they seemed attracted to the separated infant, although they also approached the unfamiliar juvenile. Thus, overall, males showed no hormonal responsiveness to mother-infant separation and avoided separated infants, while females responded hormonally to mother-infant separation and approached separated infants.

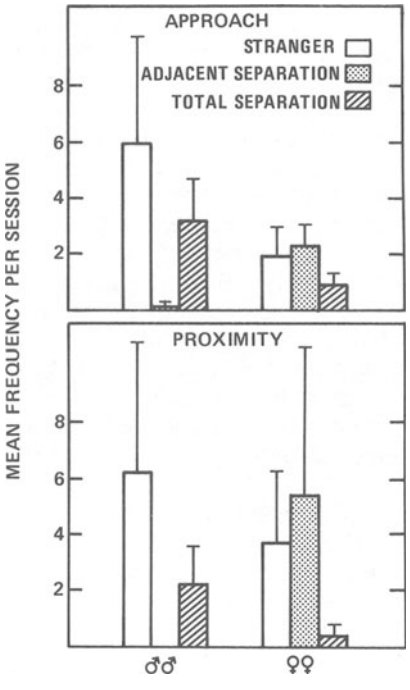



FIGURE 8

Approach and proximity scores of males and non-lactating females in adult group in response to presentation of unfamiliar conspecific (Stranger) or to mother-infant separation with mother in home cage and infant in adjacent stimulus cage (Adjacent Separation) or in another building (Total Separation). (data from Vogt et al., 1980)

Although this study indicates that adult males are not responsive to mother-infant separation, the observations that males will attack human caretakers handling infants suggest another possibility. It may be that adult male squirrel monkeys respond in a protective manner only during a period of physical threat to the infant. Also, any responsiveness to the infants could be facilitated by the males actually living with the mother-infant dyads. A second experiment examined male responsiveness under these conditions (Vogt, Siperstein, and Levine, in prep. b).

The adult males in this more recent study had lived in the same social group as the mother-infant dyads since the birth of the infants. Testing began when the infants were 4-5 months of age. Each of the 5 experimental conditions in this study involved exposing adult males for 30 minutes to a mother and/or infant separated from each other. In one of the conditions the infant was held throughout the session by an experimenter in front of the group cage which contained all other animals of the

group, including the mother. In the other 4 conditions, the mother was either in the home cage or removed to a different room while the separated infant was either placed in a small cage in front of the home cage or removed to the other room. Analysis of the blood samples obtained from the adult males at the end of each session revealed significant cortisol elevations *only* after the infant handled condition. The cortisol values for the 4 other separation conditions were unchanged from basal values. These results suggest that adult males may in fact pituitary-adrenal activation to mother-infant separation, but only during conditions in which the infant is being physically handled and therefore more clearly threatened by the experimenter. The degree to which this response is due to the males' perception of the physical threat, or to the cues emitted by the infant or the mother remains to be determined. Regardless, it does seem apparent that the physical threat to the infant, rather than the separation distress of mother and/or infant, is the more critical factor in the males' adrenocorticoid reactivity.

Females, however, in keeping with their behavioral attraction to infants during normative (Hunt et al., 1978) as well as maternal separation (Rosenblum, 1972) conditions, evidence pituitary-adrenal elevations during any sort of mother-infant separation (Vogt et al., 1980). Moreover, this adrenocorticoid reactivity to mother-infant separation was observed regardless of the endocrine changes that characterize a female during lactation. The females carrying infants and lactating, and the females without infants and not lactating, both revealed significant cortisol elevations after the separation conditions, and the patterns of significant elevations for both types of females were the same. These data contrast rather sharply with research on rodents which has indicated that only lactating females show pituitary-adrenal responsiveness to cues from disturbed pups (Smotherman, et al., 1976; Hennessy, Hollister, and Levine, in prep.). This difference may reflect the greater social complexity and cognitive ability of nonhuman primates.

In closing, we would like to note that the data reviewed in this chapter only begin to cover the various contexts in which pituitary-adrenal measures might be used to study the infant's social relations with animals other than the mother. To us, one of the more interesting possibilities for future research is to use the adrenocorticoid response to examine the development of the infant's social awareness of other group members. For instance, one might ask at what age do squirrel monkeys begin to exhibit pituitary-adrenal activation to manipulations of social group members other than the mother. At present, only intriguing hints regarding this question are available. In one study (Vogt and Levine, 1980) infants as young as 4 months of age showed a cortisol response following presentation of an unfamiliar conspecific. The mothers upon which the infants were riding did not show as clear a response. In another study, Hennessy, Vogt, Gonzales, and Levine (unpublished observations) found that under conditions of enforced proximity, infants of about 4 months of age showed pituitary-adrenal activation which may have been a response

to separation from another mother-infant dyad. These observations are, however, most preliminary. The actual extent to which hormonal measures can be used to measure such aspects of social development will need to be determined in future experiments.

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## REFERENCES

Bowden, D., Winter, P., and Ploog, D. Pregnancy and delivery behavior in the squirrel monkey *(Saimiri sciureus)* and other primates. *Folia Primatologica*, 1967, *5*, 1-42.

Bowlby, J. *Attachment and loss, vol. 2: Separation.* London: Hogarth, 1973.

Breese, G. R., Smith, R. D., Mueller, R. A., Howard, J. L., Prange, A. J., Lipton, M. A., Young, L. D., McKinney, W. E., and Lewis, J. K. Induction of adrenal catecholamine synthesizing enzymes following mother-infant separation. *Nature New Biology*, 1973, *246*, 94-96.

Coe, C. L., Mendoza, W. P., Smotherman, W. P., and Levine, S. Mother-infant attachment in the squirrel monkey: Adrenal response to separation. *Behavioral Biology*, 1978, *22*, 256-263.

Coover, G. D., Goldman, L., and Levine, S. Plasma corticosterone increases produced by extinction of operant behavior in rats. *Physiology and Behavior*, 1971, *6*, 261-263.

Friedman, S. B., and Ader, R. Adrenocortial response to novelty and noxious stimulation. *Neuroendocrinology*, 1967, *2*, 209-212.

Hanson, J. D., Larson, M. E., and Snowdon, C. T. The effects of control over high intensity noise on plasma cortisol levels in rhesus monkeys. *Behavioral Biology*, 1976, *16*, 333-340.

Harlow, H. The nature of love. *American Psychologist*, 1958, *13*, 673-685.

Harlow, H. F., and Suomi, S. J. Nature of love--simplified. *American Psychologist*, 1970, *25*, 161-168.

Harlow, F., and Zimmerman, R. R. Affectional responses in the infant monkey. *Science*, 1959, *130*, 421-431.

Hennessy, J. W., and Levine, S. The pituitary-adrenal system: A psycho-endocrine hypothesis. In J. Sprague and A. Epstein (Eds.), *Progress in psychobiology and physiological psychology (vol. 8).* New York: Academic Press, 1979.

Hennessy, M. B., Hollister, T. A., and Levine, S. Pituitary-adrenal responsiveness of mothers to pups in mice of two inbred strains. In prep.

Hennessy, M. B., Kaplan, J. N., Mendoza, S. P., Lowe, E. L., and Levine, S. Separation distress and attachment in surrogate-reared squirrel monkeys. *Physiology and Behavior*, 1979, *23*, 1017-1023.

Hill, S. D., McCormack, S. A., and Mason, W. A. Effects of artificial mothers and visual experience on adrenal responsiveness of infant monkeys. *Developmental Psychobiology*, 1973, *6*, 421-429.

Hinde, R. A. Rhesus monkey aunts. In B. M. Foss (Ed.), *Determinants of infants behavior (vol. III)*. London: Methuen, 1965.

Hrdy, S. B. Care and exploitation of nonhuman primate infants by conspecifics other than the mother. In J. S. Rosenblatt, R. A. Hinde, E. Shaw, and C. Beer (Eds.), *Advances in the study of behavior*. New York: Academic Press, 1976.

Hunt, S. M., Gamache, K. M., and Lockard, J. S. Babysitting behavior by age/sex classification in squirrel monkeys *(Saimiri sciureus)*. *Primates*, 1978, *19*, 179-186.

Itani, J. Paternal care in the wild Japanese monkey. *Primates*, 1959, *2*, 61-93.

Jensen, G. D., and Tolman, C. W. Mother-infant relationship in the monkey, *(Macaca nemestrina)*: The effect of brief separation and mother-infant specificity. *Journal of Comparative and Physiological Psychology*, 1962, *55*, 131-136.

Jones, B. C., and Clark, D. L. Mother-infant separation in squirrel monkeys living in a group. *Developmental Psychobiology*, 1973, *6*, 259-269.

Kaplan, J. The effects of separation and reunion on the behavior of mother and infant squirrel monkeys. *Developmental Psychobiology*, 1970, *3*, 43-52.

Kaplan, J. Growth and behavior of surrogate-reared squirrel monkeys. *Developmental Psychobiology*, 1974, *1*, 7-13.

Kaplan, J. Perceptual properties of attachment in surrogate-reared and mother-reared squirrel monkeys. In S. Chevalier-Skolnikoff and F. E. Poirier (Eds.), *Primate bio-social development: Biological, social, and ecological determinants*. New York: Garland Publishing, Inc., 1977.

Kaplan, J. N., Cubicciotti, D., and Redican, W. K. Olfactory discrimination of squirrel monkey mothers by their infants. *Developmental Psychology*, 1977, *10*, 447-454.

Kaplan, J. N., Cubicciotti, D. D., and Redican, W. K. Olfactory and visual differentiation of synthetically scented surrogates by infant squirrel monkeys. *Developmental Psychobiology*, 1979, *12*, 1-10.

Kaplan, J., and Russell, M. Olfactory recognition in the infant squirrel monkey. *Developmental Psychobiology*, 1974, *7*, 15-19.

Kaufman, I. C., and Rosenblum, L. A. Depression in infant monkeys separated from their mothers. *Science*, 1967, *155*, 1030-1031. (a)

Kaufman, I. C., and Rosenblum, L. A. The reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. *Psychosomatic Medicine*, 1967, *29*, 648-675. (b)

Lancaster, J. B. Play-mothering: The relations between juvenile females and young infants among free-ranging vervet monkeys *(Cercopithecus aethiops)*. *Folia Primatologia*, 1971, *15*, 161-182.

Lewis, M., and Goldberg, S. Perceptial-cognitive development in infancy: A generalized expectancy model as a function of the mother-infant interaction. *Merrill-Palmer Quarterly*, 1969, *15*, 81-100.

Mason, W. A. Wanting and knowing: A biological perspective on maternal deprivation. In E. B. Thoman (Ed.), *Origins of the infants social responsiveness*. Hillside, N.J.: Lawrence Erlbaum Associates, 1979.

Mason, W. A., and Kenney, M. D. Redirection of filial attachments in rhesus monkeys: Dogs as mother surrogates. *Science*, 1974, *183*, 1209-1211.

McKenna, J. A. Aspects of infant socialization, attachment, and maternal caretaking patterns among primates: A cross-disciplinary review. *Yearbook of Physical Anthropology*, 1979, *50*, 250-286.

Mendoza, S. P., Smotherman, W. P., Miner, M. T., Kaplan, J., and Levine, S. Pituitary-adrenal response to separation in mother and infant squirrel monkeys. *Developmental Psychobiology*, 1978, *11*, 169-175.

Meyer, J. S., Novak, M. A., Bowman, R. E., and Harlow, H. F. Behavioral and hormonal effects of attachment object separation in surrogate-peer-reared and mother-reared infant rhesus monkeys. *Developmental Psychobiology*, 1975, *8*, 425-435.

Mineka, S., and Suomi, S. J. Social separation in monkeys. *Psychological Bulletin*, 1978, *85*, 1376-1400.

Mitchell, G. D., and Brandt, E. M. Paternal behavioral in primates. In F. E. Poirier (Ed.), *Primate Socialization*. New York: Random House, 1972.

Redican, W. K. Adult male-infant interactions in nonhuman primates. In M. E. Lamb (Ed.), *The role of the father in child development*. New York: Wiley, 1976.

Reite, M., and Short, R. A. Nocturnal sleep in separated infant monkeys. *Archives of General Psychiatry*, 1978, *35*, 1247-1253.

Reite, M., Short, R., Kaufman, I. C., Stynes, A. J., and Pauley, J. D. Heart rate and body temperature in separated monkey infants. *Biological Psychiatry*, 1978, *13*, 91-105. (a)

Reite, M., Short, R., and Seiler, C. Physiological correlates of maternal separation in surrogate-reared infants: A study in altered attachment bonds. *Developmental Psychobiology*, 1978, *11*, 427-435. (b)

Rosenblum, L. A. Mother-infant relations and early behavioral development in the squirrel monkey. In L. A. Rosenblum and R. W. Cooper (Eds.), *The squirrel monkey*. New York: Academic Press, 1968.

Rosenblum, L. A. Infant attachment in monkeys. In R. Schaffer (Ed.), *The origins of human social relations*. New York: Academic Press, 1971. (a)

Rosenblum, L. A. The ontogeny of mother-infant relations in macaques. In H. Moltz (Ed.), *The ontogeny of vertebrate behavior*. New York: Academic Press, 1971. (b)

Rosenblum, L. A. Sex and age differences in response to infant squirrel monkeys. *Brain Behavior and Evolution*, 1972, *5*, 30-40.

Rosenblum, L. A. Sex differences, environmental complexity, and mother-infant relations. *Archives of Sexual Behavior*, 1974, *2*, 117-128.

Rosenblum, L. A., and Coe, C. L. The influence of social structure on squirrel monkey socialization. In S. Chevalier-Skolnikoff and F. E. Poirier (Eds.), *Primate bio-social development.* New York: Garland, 1977.

Rosenblum, L. A., and Kaufman, I. C. Variations in infant development and response to maternal loss in monkeys. *American Journal of Orthopsychiatry,* 1968, *38,* 418-426.

Sackett, G. P. Unlearned responses, differential recurring experiences and the development of social attachments by rhesus monkeys. In L. A. Rosenblum (Ed.), *Primate behavior (vol. 1).* New York: Academic Press, 1970.

Sackett, G. P., and Ruppenthal, G. C. Some factors influencing the attraction of adult female macaque monkeys to neonates. In M. Lewis and L. A. Rosenblum (Eds.), *The effect of the infant on its caregiver.* New York: Wiley, 1974.

Seay, B., Hansen, E., and Harlow, H. F. Mother-infant separation in monkeys. *Journal of Child Psychology and Psychiatry,* 1962, *3,* 123-132.

Smotherman, W. P., Hunt, L. E., McGinnis, L. M., and Levine, S. Mother-infant separation in group-living rhesus macaques: A hormonal analysis. *Developmental Psychobiology,* 1979, *12,* 211-217.

Smotherman, W. P., Wiener, S. G., Mendoza, S. P., and Levine, S. Pituitary-adrenal responsiveness of rat mothers to noxious stimuli and stimuli produced by rat pups. In K. Elliott and D. W. Fitzsimons (Eds.), *Breast-feeding and the mother.* Amsterdam: Elsevier, 1976.

Takeshita, H. On the delivery behavior of squirrel monkeys *(Saimiri sciurea)* and a mona monkey *(Cercopithecus mona). Primates,* 1961-1962, *3,* 59-72.

Vogt, J. L., Carlson, H., and Menzel, E. Social behavior of a marmoset *(Saguinus fuscicollis)* group I: Parental care and infant development. *Primates,* 1978, *19,* 715-726.

Vogt, J. L., Coe, C. L., Lowe, E., and Levine, S. Behavioral and pituitary-adrenal response of adult squirrel monkeys to mother-infant separation. *Psychoneuroendocrinology,* 1980, in press.

Vogt, J. L., and Levine, S. Response of mother and infant squirrel monkeys to separation and disturbance. *Physiology and Behavior,* 1980, *24,* 829-832.

Vogt, J. L., Lowe, E., and Levine, S. Mother-infant separation in the squirrel monkey: Social influences upon behavioral and adreno-corticoid responses. In prep. (a)

Vogt, J. L., Siperstein, L., and Levine, S. Response of adult male squirrel monkeys to mother-infant separation. In prep. (b)

Watson, J. S., and Ramey, C. T. Reactions to response-contingent stimulation in early infancy. *Merrill-Palmer Quarterly,* 1972, *18,* 219-227.

Wilson, E. O. *Sociobiology: The new synthesis.* Cambridge, MA: Harvard University Press, 1975.

Wood, B. S., Mason, W. A., and Kenney, M. D. Contrasts in visual responsiveness and emotional arousal between rhesus monkeys raised with living and those raised with inanimate substitute mothers. *Journal of Comparative and Physiological Psychology*, 1979, *93*, 368-377.